$$F^{(q)}_{3,n}(x) = r^q\, p^{(n,n)}_q\left(\frac{x_3}{r}\right)$$

$|r|^2 = |x_1|^2 + |x_2|^2 + |x_3|^2$

(3.37)

$$\partial_x\left(F^{(q)}_{3,n}(x)\, p_n(x_1,x_2)\right) = \partial_x\left(F^{(q)}_{3,n}(x)\right) p_n$$

$$= \left(e_1\frac{\partial}{\partial x_1}F^{(q)}_{3,n}(x) + e_2\frac{\partial}{\partial x_2}F^{(q)}_{3,n}(x) + e_3\frac{\partial}{\partial x_3}F^{(q)}_{3,n}(x)\right) p_n$$

$$= \left[e_1\frac{\partial}{\partial x_1}\left(r^q\, p^{(n,n)}_q\left(\frac{x_3}{r}\right)\right) + e_2\frac{\partial}{\partial x_2}\left(r^q\, p^{(n,n)}_q\left(\frac{x_3}{r}\right)\right) + e_3\frac{\partial}{\partial x_3}\left(r^q\, p^{(n,n)}_q\left(\frac{x_3}{r}\right)\right)\right] p_n$$

$r = \sqrt{x_1^2+x_2^2+x_3^2}$

$$= \left[e_1\, q\,(r^2)^{\frac{q}{2}-1} x_1\, p\left(\frac{x_3}{r}\right) + e_1 r^q\, r^2\, p'\left(\frac{x_3}{r}\right)\times\left(\frac{-\frac{1}{r}x_1\frac{x_3}{r}}{r^2}\right)\right.$$

r^{q-2} $\quad$ $r^{q-2}\cdot r^2$

$$\left. + e_2\, q\, r^{q-2} x_2\, p\left(\frac{x_3}{r}\right) + e_2 r^q\, p'\left(\frac{x_3}{r}\right)\times\left(\frac{-\frac{x_2}{r}x_3}{r^2}\right)\right.$$

$$+ e_3 \, q \, r^{q-2} n_3 \, p\left(\frac{n_3}{r}\right) + e_3 \, r^q \, p'\left(\frac{n_3}{r}\right) \times \left(\frac{n - \frac{n_3}{r} n_3}{r^2}\right) \Bigg] p_n$$

$$= r^{q-2} \Big[q \, p\left(\frac{n_3}{r}\right) n_1 e_1 - e_1 \left(-\frac{n_3}{r}\right) p'\left(\frac{n_3}{r}\right) n_1 e_1 \Big]$$

$$+ r^{q-2} \Big[q \, p\left(\frac{n_3}{r}\right) n_2 e_2 - e_2 \left(-\frac{n_3}{r}\right) p'\left(\frac{n_3}{r}\right) n_2 e_2 \Big]$$

$$+ e_3 q \, r^{q-2} n_3 \, p\left(\frac{n_3}{r}\right) + e_3 r^q \, p'\left(\frac{n_3}{r}\right) \times \frac{\left(1 - \frac{n_3^2}{r^2}\right)}{r} \Bigg] p_n$$

$$= r^q \Big[q \, p\left(\frac{n_3}{r}\right) - \frac{n_3}{r} p'\left(\frac{n_3}{r}\right) \Big]$$

$$(e_1 n_1 + e_2 n_2) p_n$$

$$+ e_3 r^{q-1} \left[q \frac{n_3}{r} p\left(\frac{n_3}{r}\right) + \left(1 - \left(\frac{n_3}{r}\right)^2\right) p'\left(\frac{n_3}{r}\right) \right] p_n$$

$$\Rightarrow \partial F^{(q)}_{3,n} = r^{q-2} \left[q\, p^{(n,n)}_q \left(\frac{n_3}{r}\right) - \frac{n_3}{r} p^{(n,n)'}_q \left(\frac{n_3}{r}\right) \right] (e_1 n_1 + e_2 n_2) p_n$$

$$+ r^{q-1} \left[q \frac{n_3}{r} p^{(n,n)}_q \left(\frac{n_3}{r}\right) + \left(1 - \left(\frac{n_3}{r}\right)^2\right) p^{(n,n)'}_q \left(\frac{n_3}{r}\right) \right] p_n$$